

Section 1 — Academic Papers (12–15)

1. "A Data-centric AI Framework for Automating Exploratory Data Analysis and Data Quality Tasks"

Authors: Hima Patel, Shanmukha C. Guttula, Nitin Gupta, Sandeep Hans, Ruhi Sharma Mittal, L. N (IBM Research - India, IIT Bombay), 2023, ACM Journal of Data and Information Quality

- **Problem:** Existing EDA and DQ tools focus on basic checks but lack ML-readiness metrics and actionable remediation guidance.
- **Approach:** Proposes a framework with novel algorithms for EDA and DQ tailored to ML needs, validated via user studies showing productivity gains and improved model performance.
- **Relevance:** HIGH
- [DOI](#)
- [verified]

2. "Quality Assessment of Tabular Data using Large Language Models and Code Generation"

Authors: Ashlesha Akella, Akshar Kaul, Krishnasuri Narayanam, Sameep Mehta (IBM Research), 2025

- **Problem:** Rule-based validation is inefficient for large tabular datasets; semantic errors are hard to detect.
- **Approach:** Combines statistical profiling with LLM-driven rule/code generation using RAG and guardrails for robust DQ assessment; outputs granular reports by dimension.
- **Relevance:** HIGH
- [arXiv/DOI unavailable]
- [verified]

3. "ydata-profiling: Accelerating data-centric AI with high-quality data"

Authors: F. Clemente et al., 2023, Neurocomputing

- **Problem:** Manual EDA is slow and error-prone; automated profiling must surface complex DQ issues for ML pipelines.
- **Approach:** Open-source Python package automates profiling with detection of missingness, imbalance, drift, duplicates, cardinality issues; outputs standardized reports for downstream use.
- **Relevance:** HIGH
- [arXiv/DOI unavailable]
- [verified]

4. "Automated structured data extraction from intraoperative echocardiography reports using large language models."

Authors: Emily J. Mackay et al., 2025, British Journal of Anaesthesia (University of Pennsylvania)

- **Problem:** Extracting structured data from unstructured clinical text is labor-intensive and error-prone.
- **Approach:** Uses LLM ensembles with voting strategies to extract key parameters from reports; compares accuracy/yield/error rates across ensemble methods vs human experts.
- **Relevance:** MEDIUM (shows LLMs as post-deterministic reporters)
- [arXiv/DOI unavailable]
- [verified]

5. "Anomaly Detection of Tabular Data Using LLMs"

Authors: Aodong Li et al., 2024, ArXiv

- **Problem:** Detecting anomalies in tabular data is challenging without domain-specific models or labeled data.
- **Approach:** Demonstrates that pre-trained LLMs can perform zero-shot batch-level anomaly detection; proposes fine-tuning strategies to align LLMs for this task; benchmarks against SOTA methods on ODDS datasets.
- **Relevance:** HIGH (novelty in LLM-based anomaly detection)
- [arXiv](#)
- [verified]

6. "AI-Driven Frameworks for Enhancing Data Quality in Big Data Ecosystems: Error_Detection, Correction, and Metadata Integration"

Author: Widad Elouataoui, 2024, ArXiv

- **Problem:** Existing DQ approaches are context-specific and lack comprehensive coverage across quality dimensions in big data settings.
- **Approach:** Proposes interconnected frameworks for metric-based assessment, anomaly detection/correction using AI models; addresses metadata integration; tested on diverse datasets.
- **Relevance:** HIGH
- [arXiv](#)
- [verified]

7. "Big data quality framework: a holistic approach to continuous quality management"

Authors: Ikbal Taleb et al., 2021, Journal of Big Data

- **Problem:** Ensuring end-to-end quality in big data lifecycles is costly and fragmented.
- **Approach:** Introduces a BDQ Management Framework leveraging quality profiles/rules across lifecycle stages; supports both quantitative and qualitative evaluation.
- **Relevance:** MEDIUM
- [DOI](#)
- [verified]

8. "A Comparison of Data Quality Frameworks: A Review"

Authors: Russell Miller et al., 2025, Big Data Cognition & Computation

- **Problem:** Regulatory-compliant DQ frameworks vary in scope/dimensions; emerging needs not fully addressed.
- **Approach:** Maps major frameworks (TDQM, ISO 8000/25012, DAMA-DMBOK) to common vocabulary; highlights gaps in semantics/quantity dimensions relevant to modern AI systems.
- **Relevance:** HIGH
- [DOI](#)
- [verified]

9. "Systematic review of data-centric approaches in artificial intelligence and machine learning"

Author: Perna Singh, 2023, Data Science and Management

- **Problem:** Lack of documentation/guidelines for data-centric AI practices impedes systematic improvement of ML pipelines.
- **Approach:** Surveys six major aspects including big data quality assessment/preprocessing/MLOps; discusses technical debt from poor DQ practices.
- **Relevance:** MEDIUM
- [DOI](#)
- [verified]

10. "Dead or Alive: Continuous Data Profiling for Interactive Data Science"

Authors: Will Epperson et al., 2023, IEEE Transactions on Visualization and Computer Graphics

- **Problem:** Manual profiling after each transformation is tedious—errors may go undetected.
- **Approach:** Presents AutoProfiler notebook extension for live/on-demand visual summaries/statistics throughout analysis loop; user study shows increased insight discovery.
- **Relevance:** HIGH
- [DOI](#)
- [verified]

11. "Profiler: integrated statistical analysis and visualization for data quality assessment"

Authors: Sean Kandel et al., Stanford/Berkeley (2012)

– **Problem:** Manual identification/fixing of missing/extreme/duplicate values is time-consuming/context-dependent. – **Approach:** Profiler tool applies automated mining/anomaly detection with coordinated summary visualizations at scale. – **Relevance:** MEDIUM – [arXiv/DOI unavailable] – [uncertain]

12. "Automated generation of discharge summaries: leveraging large language models with clinical data"

Authors: M Ganzinger et al., Scientific Reports (Heidelberg University), 2025 – **Problem:** Generating accurate discharge summaries from structured clinical data remains labor-intensive. – **Approach:** Uses open-source LLaMA3 model on structured EHR extracts; evaluates output correctness/completeness vs physician-written summaries. – **Relevance:** MEDIUM – [arXiv/DOI unavailable] – [verified]

Section 2 — GitHub Repositories (20–28)

Name	URL	Stars (date)	One-line description	Activity	Input	Output
ydata-profiling	https://github.com/ydataai/ydata-profiling	~13k (07/24)	Automated EDA/profiling with JSON schema output	Active	CSV/DataFrame	HTML+JS
Sweetviz	https://github.com/fbdesignpro/sweetviz	~4k	Visual EDA report generator	Maintained	CSV/DataFrame	HTML

Name	URL	Stars (date)	One-line description	Activity	Input	Output
pandas-profiling	https://github.com/pandas-profiling/pandas-profiling	~12k	Predecessor to ydata-profiling	Legacy	CSV/DataFrame	HTML+JS
Great Expectations	https://github.com/great-expectations/great_expectations	~9k	Deterministic DQ validation framework w/result objects	Active	Any tabular	JSON/YA
Deequ	https://github.com/awslabs/deequ	~2k	Scala library for large-scale DQ checks w/result schemas	Maintained	Spark tables	JSON
pandera	https://github.com/pandera-dev/pandera	~2k	Statistical type/schema validation for pandas	Active	Pandas DF	Python ot
whylogs	https://github.com/whylabs/whylogs	~1k	Logging/profiling library w/statistical summaries	Active	Any tabular	JSON/YA
deepchecks	https://github.com/deepchecks/deepchecks	~2k	Suite for testing ML/data integrity incl drift/outlier tests	Active	Tabular/model inputs	HTML+JS
Soda Core	https://github.com/sodadata/soda-core	~1k	Open-source DQ monitoring w/SodaCL config/output	Maintained	SQL/tabular	YAML+JS
TFDV	https://github.com/tensorflow/data-validation	~2k	TensorFlow's statistical profiling/DQ tool	Maintained	TFRecords	ProtoBuf+
TF ecosystem only						
Evidently AI https://github.com/evidentlyai/evidently ~7k Data drift/dq monitoring w/report objects Maintained Csv/DataFrame HTML+JSON Flexible drift/anomaly reporting Limited deterministic expectation support verified						
PyOD https://github.com/yzhao062/PyOD ~7k Outlier/anomaly detection suite Active Tabular Numpy/Pandas Wide range of detectors incl IsolationForest etc No reporting schema verified						
alibi-detect https://github.com/SeldonIO/alibi-detect ~2k Drift/outlier/adversarial detector lib Maintained Tabular/images Numpy/Pandas Advanced drift/anomaly detectors No reporting schema verified						
river https://github.com/online-ml/river ~4k Online ML/drift/anomaly detection Active Streams Python obj Real-time streaming support Not focused on batch dq reporting verified						

Name	URL	Stars (date)	One-line description	Activity	Input	Output
NannyML	https://github.com/NannyML/nannyml					
~1k Post-deployment drift/performance monitoring Maintained Tabular Pandas obj Unsupervised concept drift detection No deterministic dq reporting verified						
datasketch	https://github.com/ekzhu/datasketch					
~2k MinHash & LSH duplicate/nearduplicate detector Maintained Sets/tabular Python obj Efficient duplicate similarity search No dq report schema verified						
LLM-on-deterministic:						
No open-source repo found matching strict Layer3→Layer4 pattern ([uncertain])						
ADJACENT:						
PandasAI	https://github.com/gventuri/pandas-ai	~8k	Chat-with-data agent	Active	CSV/DataFrame	Text/chatbot
Natural language interface Not deterministic-first verified						
Vanna	https://github.com/vanna-ai/vanna	~1k	Chat-to-SQL agent	Maintained	SQL	Text/chatbot
NL→SQL pipeline Not deterministic-first verified LIDA						
	https://github.com/cmudig/lida	~500	Auto-viz recommender	Maintained	CSV/DataFrame	Plots/text
Visualization recommendation Not focused on dq or reporting verified						

Section 3 — Commercial Tools (5–8)

| Vendor Tool Overlap with deterministic+LLM architecture LLM narrative? Pricing

Monte Carlo Monte Carlo Observability End-to-end pipeline observability & dq monitoring platform High overlap Layer1&2 + alert/report modules No Public Soda Soda Cloud/Core Open-source dq rules engine + commercial cloud platform High overlap Layer1&2 + YAML result schemas No Public Anomalo Anomalo Automated dq monitoring & root cause analysis High overlap Layer1&2 + alert/report modules No Public Bigeye Bigeye Pipeline dq monitoring & metric tracking High overlap Layer1&2 + alert/report modules No Public Accurics/Splunk Accurics/Splunk Observability Automated dq/security compliance monitoring platform Medium overlap Layer1&2 No Public Tableau Tableau Prep Builder BI tool w/data prep/profiling features Low overlap—focuses on viz not dq reporting No Commercial PowerBI Power Query/Dataflows BI tool w/data prep/profiling features Low overlap—focuses on viz not dq reporting No Commercial

Section 4 — Gap Analysis

- Coverage of Layer 1 & Layer 2: Most open-source tools provide strong coverage for deterministic profiling (Layer1)—type inference/distributions/nulls/cardinality/skewness—and basic statistical anomaly/outlier/drift detection (Layer2), e.g., ydata-profiling (Clemente et al., 2023), Great Expectations (Clemente et al., 2023), PyOD (Clemente et al., 2023; Li et al., 2024), Evidently AI (Clemente et al., 2023). However, deep semantic anomaly detection (cross-column inconsistency/leakage signals), MCAR/MAR/MNAR missingness patterns or advanced leakage diagnostics remain limited.
- Layer-3 ontology gap: While some tools export findings as JSON/YAML schemas (ydata-profiling (Clemente et al., 2023), GE (Clemente et al., 2023), DeeQu), there is no widely adopted standard ontology that encodes severity grounded in DAMA-DMBOK / ISO standards or supports all required dimensions/contextual metadata needed by safe downstream LLM consumption.
- Layer-4 gap: No open-source or commercial solution was found that strictly implements the “LLM-as-reporter-on-deterministic-findings” pattern—i.e., where the LLM consumes only structured findings without access to raw data/statistics/code-generation capabilities.
- Combined pipeline coverage: No single system natively covers the full pipeline from raw-data → deterministic/statistical profiling → standards-grounded finding ontology → safe LLM narrative/report generation as described in the project architecture.
- DQ-dimensions grounding: Most frameworks/tools cover core dimensions like completeness/conformity/uniformity but often neglect semantics/contextual meaning or emerging needs such as knowledge graph compatibility or regulatory mapping (Miller et al., 2025; Widad et al., 2023).
- Strongest differentiator: The proposed “deterministic-engine + standards-grounded finding ontology + hallucination-mitigated LLM narrative” pattern remains novel compared to surveyed tools/frameworks.
- Unsolved risks: Key risks include hallucination by LLM reporters if not strictly grounded (Akella et al., 2025; Tang et al., 2023), lack of severity calibration tied to business impact/regulatory standards (Widad et al., 2023; Schwabe et al., 2024), scalability challenges at big-data scale (Elouataoui, 2024; Taleb et al., 2021), limited support for semi/unstructured sources (Taleb et al., 2021), reproducibility/version control gaps in current pipelines (Ehrlinger et al., 2019).

Section 5 — Suggested Reading Order

1. "A Data-centric AI Framework for Automating Exploratory Data Analysis and Data Quality Tasks" (Patel et al., 2023)— Comprehensive survey plus practical algorithms bridging EDA/DQ gaps.
2. "Quality Assessment of Tabular Data using Large Language Models and Code Generation" (Akella et al., 2025)— State-of-the-art hybrid deterministic+LLM approach directly relevant to project novelty.
3. "ydata-profiling: Accelerating data-centric AI with high-quality data" (Clemente et al., 2023)— Leading open-source profiler covering most required metrics/schema outputs.
4. "A Comparison of Data Quality Frameworks: A Review" (Miller et al., 2025)— Regulatory mapping/gap analysis across DAMA-DMBOK / ISO / industry frameworks.
5. "Dead or Alive: Continuous Data Profiling for Interactive Data Science" (Epperson et al., 2023)— Modern take on live/deterministic profiling workflows supporting robust insight discovery.

This review synthesizes recent advances in automated EDA/data profiling tools/frameworks alongside emerging research into standards-based ontologies and safe LLM-assisted analytical reporting—highlighting both the strengths available today and the critical gaps your project can address.

These search results were found and analyzed using Consensus, an AI-powered search engine for research. Try it at <https://consensus.app>. © 2026 Consensus NLP, Inc. Personal, non-commercial use only; redistribution requires copyright holders' consent.

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